Odette School of Business

BSMM- 8730: Data Acquisition and Management

D.4.1 User Requirements and Analysis Document

Global Child Mortality Visual Analytics Application (2010–2024)

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1. Introduction and Scope

1.1. Introduction

The Global Child Mortality Visual Analytics application's main aim is to analyze mortality patterns between 2010 and 2024 (Amegbor & Addae, 2023). The application presents this data through dashboards, charts, maps, and statistical summaries so users can easily understand the differences. The following indicators will be used: healthcare access, income levels, sanitation, and maternal health. The data will first be downloaded and then cleaned in Excel (Microsoft Excel, 2024), where duplicates and blanks will be removed. The application aims to support evidence-based decisionmaking for data exploration rather than building causal relationships among medical diagnosis. The application will use Microsoft Excel for the initial examination of raw datasets, preliminary validation, and basic data-quality checks before the data are imported into R. R will serve as the primary tool for data cleaning, transformation, integration, and statistical analysis. MySQL will be used to store and manage the clean and integrated datasets, while Microsoft Power BI will be used to create interactive dashboards and visualizations.

1.2. Objectives

The main objectives for the project are -

- Understanding and analyzing the mortality trends from 2010 to 2024 (Child mortality rate, 2026).
- Comparison among different countries and regions for the mortality rates
- Exploring the main relationship between healthcare indicators and mortality
- Using Power BI to create visuals that help users interact with filters and examine data more effectively (Microsoft Power BI, 2024).
- Showing and presenting all the findings regarding education, research, and public health analysis

1.3. Scope

The project covers the collection, cleaning, integration, and analysis of global child mortality data from 2010–2024 using international datasets. Excel will be used for initial data checks, R for data cleaning and statistical analysis, MySQL for data storage, and Power BI for interactive dashboards. The dashboard will include a world map, a year slider, filters for country, region, and income group, and a scatter plot comparing immunization coverage with child mortality when data is available. The analysis will include descriptive statistics, correlation, regression, and time-series analysis where possible. The project excludes individual patient records, real-time monitoring, and medical diagnosis

2. Stakeholder Roles

Stakeholder	**Role and requirements**
Researchers and analysts	Analyze trending mortality data, compare different countries, and investigate relationships between indicators. (UNICEF et al., 2025).
Public health professionals	Help identify areas that require more investigation, and research and find geographic differences
Students and educators	By using the Power BI visuals, they will be able to understand the global child mortality patterns
General users	Without having technical knowledge, they will be able to interpret the findings, view summaries, and select different countries
Project developer	Collect all the data, then analyze it; use Excel to clean and further process the data. Using Power BI will make the data more efficient by adding visualizations, testing, and documentation.
Data providers	Data will be provided for analysis, training, and processing.

3. Functional Requirements

F1	The application should be able to load all the data between 2010- 2024 that will be selected for global child mortality datasets	H
F2	The application should be able to handle data cleaning by using different techniques like identifying missing values, removing duplicate records, and inconsistent formats	H

F3	The application should have attractive dashboard visuals, including mortality indicators and summary stats.	H
F4	The application should show a modern, trending dashboard display that includes mortality trends.	H
F5	The application should allow users to filter data by country, geographical region, and year.	H
F6	The application can show geographical visualizations.	H
F7	The application should include clear headings, unit labels, data sources, and explanations.	H
F8	The application should show exploratory data analysis identified by the patterns and show summaries	M
F9	The application should support user data export.	M
F10	The application should raise appropriate alerts if any data or filter is missing.	H
F11	All the data will be stored in MYSQL with a proper schema.	H

4. Non-Functional Requirements

ID	Requirement	Expected outcome
NF1	Usability	The application interface should be easily readable and understandable by the users and should have a consistent navbar, attractive charts, and proper filter controls
NF2	Performance	All dashboard and filter changes should be possible within 3 seconds, with defined testing conditions.
NF3	Data integrity	Properly consistent indicators, valid observations, data transformation, and calculations should be accurate and valid
NF4	Reliability	Without crashing, the application should filter proper data
NF5	Maintainability	Proper documentation, data logic, and visuals should be organized so that errors and future updates can be managed efficiently.
NF6	Compatibility	The application should be compatible and work on all browsers
NF7	Security	The application should include a security layer to prevent sensitive information leaks and ensure all data is secure.
NF8	Accessibility	Proper headings, descriptions of charts, and information should be included
NF9	Scalability	If data is available, then the application should show data and filters.

5. Acceptance Criteria

ID	Acceptance criterion	Success condition
AC-01	Data coverage	Data from 2010-2024 is used in the app.
AC-02	Data quality	All duplicate values, records, and missing values are processed and filtered out.
AC-03	Country comparison	In the app, users can filter by country and region.
AC-04	Interactive filters	Selecting filters updates the visualizations. Filters include country, region, and year.
AC-05	Geographical analysis	The relevant country-level mortality data is appropriately displayed.
AC-06	Summary dashboard	Correct data is expected, including child mortality and stats
AC-07	World map	Data regarding the world map should be displayed using filters
AC-08	Interactive filters	In the dashboard, create creative and interactive filters with a year and country
AC-09	Comparison	When data is available, it should be displayed in a scatter plot
AC-10	Stats analysis	Planned data after analysis should be displayed

AC-11	Data store	Properly cleaned and organized data stored in MySQL

6. Revision History

This is the user requirement section. Before the project proposal was developed and authorized, there were no significant updates. Given the user and functional requirements, which will be developed using Excel, R, MySQL, and Power BI, the strategy suggested in that document was carried out.

Version	Date	Description	Status
D1	27/9/2026	Project pre-proposal	Completed
D.2.1	2/10/2026	Full project proposal	Completed
D.4.1	23/10/2026	User requirements	Submitted

References

Data Analytics Tools

Microsoft Corporation. (2024). Microsoft Power BI [Computer software]. Microsoft. https://www.microsoft.com/en-us/power-platform/products/power-bi

Microsoft Corporation. (2024). Microsoft Excel [Computer software]. Microsoft. https://www.microsoft.com/en-us/microsoft-365/excel

Datasets

Child mortality rate. (2026, March 17). Our World in Data. https://ourworldindata.org/grapher/child-mortality-igme

Other References

Amegbor, P. M., & Addae, A. (2023). Spatiotemporal analysis of the effect of global development indicators on child mortality. International Journal of Health Geographics, 22(1), 9.

UNICEF, World Health Organization, World Bank Group, & United Nations Department of Economic and Social Affairs, Population Division. (2025). Levels & trends in child mortality: Report 2024. UNICEF. https://data.unicef.org/resources/levels-and-trends-in-child-mortality-2024

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